

Technical Data Sheet

Industrial Hydraulic Pump — Model HP-4000

HydroDyn Pumps Pvt. Ltd.

1. Product Overview

The HP-4000 is a high-performance industrial hydraulic pump designed for demanding fluid power applications. It delivers reliable, continuous-duty performance in manufacturing, mining, and heavy machinery installations. Model number: HP-4000. SKU: HDPUMP-4000-230V.

2. Electrical Specifications

The HP-4000 operates on a standard three-phase industrial supply. Rated voltage: 230 V (±10%). Rated current: 16 A. Frequency: 50 Hz. Power factor: 0.87. The motor is IE3 efficiency class compliant.

3. Performance Specifications

Rated power output: 5 HP (3.73 kW). Maximum operating pressure: 250 bar (3,626 psi). Nominal flow rate: 45 L/min at rated speed. Pump speed: 1450 RPM at 50 Hz. Volumetric efficiency: ≥ 92% at rated conditions.

4. Physical Specifications

Net weight: 38 kg (83.8 lb). Overall dimensions (L × W × H): 580 mm × 280 mm × 320 mm. Fluid port connections: SAE flange, 1-inch BSP. Mounting: Four-bolt flange mount, B3/B5 IEC.

5. Operating Conditions

Operating temperature range: −10 °C to +70 °C. Storage temperature: −20 °C to +85 °C. Recommended hydraulic fluid: ISO VG 46 mineral oil. Filtration: 25 µm nominal. Protection class: IP55.

6. Consolidated Specifications Table

Parameter	Value	Unit	Notes
Rated Voltage	230	V	±10%, 3-phase
Rated Current	16	A	Full load
Rated Power	5 HP / 3.73 kW	HP / kW	IE3 motor
Max Pressure	250	bar	3,626 psi equivalent
Flow Rate	45	L/min	At rated speed
Pump Speed	1450	RPM	50 Hz supply
Net Weight	38	kg	83.8 lb equivalent

Length	580	mm	22.8 inch
Width	280	mm	11.0 inch
Height	320	mm	12.6 inch
Min Operating Temp	-10	°C	14 °F
Max Operating Temp	70	°C	158 °F
Protection Class	IP55	—	IEC 60529

7. Certifications & Compliance

CE Marked — EU Machinery Directive 2006/42/EC. ISO 4413 — Hydraulic fluid power safety requirements. IEC 60034-30-1 — IE3 motor efficiency class. RoHS compliant. Country of origin: India.

8. Application Range

Suitable for: industrial presses, CNC machine tools, injection moulding machines, test rigs, and automated manufacturing equipment. Not suitable for: explosive atmospheres (ATEX-rated versions available on request), water-glycol fluids without special seals, or continuous operation above 70 °C.

■ DEMO / SYNTHETIC DATA — For IPTE hackathon demonstration only. Specifications are representative but not sourced from a real product.